

Introduction to Transient Analysis with the use of Laplace Transform

Introduction to Transient Analysis

COMMUTATION, INITIAL CONDITIONS

Commutation may be the cause of abrupt changes of current or voltage in circuit. However, constraints exist, which apply to circuit. They result from the fact that instantaneous power in real circuits cannot be infinitely great.

$$p(t) = \frac{dW(t)}{dt} < \infty$$

It means continuity of the energy function - that continuity must be preserved during the time of commutation also

Introduction to Transient Analysis

COMMUTATION, INITIAL CONDITIONS

Basing on the law of energy continuity in circuit and knowing that the value of accumulated, the energy:

in magnetic field of inductor with inductance L , through which flows current i_L is given by:

$$W_L(t) = \frac{1}{2} L i_L^2(t)$$

FIRST LAW OF COMMUTATION

The current flowing through the coil must not change abruptly, which means that the coil current at the moment immediately after commutation is equal to the current just before commutation

$$i_L(0^+) = i_L(0^-)$$

in electric field of capacitor with capacity C , charged to voltage u_C is given by:

$$W_C(t) = \frac{1}{2} C u_C^2(t)$$

SECOND LAW OF COMMUTATION

The voltage on the capacitor cannot be changed abruptly, which means that the voltage on the capacitor just after commutation is equal to the voltage just before commutation

$$u_C(0^+) = u_C(0^-)$$

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COMMUTATION, INITIAL CONDITIONS

There are no reasons to exclude abrupt changes of other quantities in the circuit i.e., coil voltages u_L , capacitor currents i_C , or resistor currents and voltages u_R, i_R .

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COMMUTATION, INITIAL CONDITIONS

Assuming that we will consider the commutation moment as the starting moment ($t_k = 0$) of transient circuit analysis, it is important to determine the initial conditions of the process.

Initial conditions are a set of values of currents in inductances and voltages on the capacitances of the system at the initial moment. Initial conditions determine the total value of energy stored in the system at the moment $t_k = 0$.

Determining the initial conditions in the circuit involves:

- resolving the steady state of the circuit before commutation, determining the time form of this solution for the time of commutation. This means that the basis for determining the initial conditions of the circuit are commutation laws.
- determining the time form of this solution for the time of commutation,
- determining the solution corresponding to the moment of commutation.

This means that the basis for determining the initial conditions of the circuit are commutation laws.

Initial conditions can be (and often are) equal to zero.

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LAPLACE TRANSFORM

In the analysis of practical electronic circuits we consider functions $f(t)$ of real variable t satisfying the following conditions:

- function $f(t)$ is defined for $t > 0$ and equal zero and when $t < 0$,
- absolute value of function $f(t)$ would not raise faster than exponential function, when $t \rightarrow \infty$ ($f(t) \leq Me^{bt}$, where $M > 0$ and $b > 0$)

The transformation that assigns the function $f(t)$ to the real variable t , the function $F(s)$ which is the function of the complex variable $s = \sigma + j\omega$ by the relationship:

$$L[f(t)] = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

we call direct (forward) **Laplace transformation** or **L-transformation**

We define the function $F(s)$ of complex variable s as transform of function $f(t)$.

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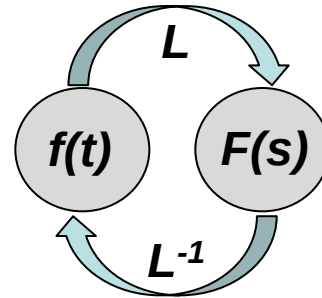
LAPLACE TRANSFORM

Determining the function $f(t)$ (called the original function) corresponding to the known function $F(s)$ allows **inverse (backward) Laplace transform** also called **L^{-1} -transform**:

$$f(t) = L^{-1}[F(s)] = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s) e^{-st} ds$$

For a given Laplace transform only one signal in time domain exists, what means that:

$$L^{-1}\{L[f(t)]\} = f(t) \quad \text{for } t > 0$$



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LAPLACE TRANSFORM OF CHOSEN FUNCTIONS

$$f(t) = \delta(t)$$

$$F(s) = 1$$

$$f(t) = \mathbf{1}$$

$$F(s) = \frac{1}{s}$$

$$f(t) = \mathbf{1}a$$

$$F(s) = \frac{a}{s}$$

$$f(t) = \mathbf{1}t$$

$$F(s) = \frac{1}{s^2}$$

$$f(t) = \mathbf{1}t^n$$

$$F(s) = \frac{n!}{s^{n+1}}$$

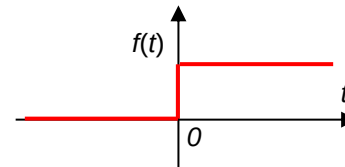
$$f(t) = \mathbf{1}e^{-at}$$

$$F(s) = \frac{1}{s+a}$$

$$f(t) = \mathbf{1}\frac{1}{a}(1 - e^{-at})$$

$$F(s) = \frac{1}{s(s+a)}$$

$\mathbf{1}(t)$ – unit step function



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LAPLACE TRANSFORM OF CHOSEN FUNCTIONS

$$f(t) = \mathbf{1}te^{-at} \qquad F(s) = \frac{1}{(s+a)^2}$$

$$f(t) = \mathbf{1}t^n e^{-at} \qquad F(s) = \frac{n!}{(s+a)^{n+1}}$$

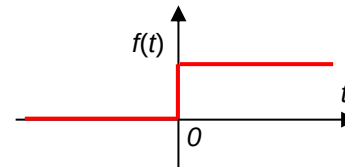
$$f(t) = \mathbf{1}\sin(\omega t) \qquad F(s) = \frac{\omega}{s^2 + \omega^2}$$

$$f(t) = \mathbf{1}\cos(\omega t) \qquad F(s) = \frac{s}{s^2 + \omega^2}$$

$$f(t) = \mathbf{1}e^{-at} \sin(\omega t) \qquad F(s) = \frac{\omega}{(s+a)^2 + \omega^2}$$

$$f(t) = \mathbf{1}e^{-at} \cos(\omega t) \qquad F(s) = \frac{s+a}{(s+a)^2 + \omega^2}$$

$\mathbf{1}(t)$ – unit step function



Introduction to Transient Analysis

BASIC PROPERTIES OF LAPLACE TRANSFORM LINEARITY

If functions $f_1(t)$ and $f_2(t)$ has transforms i.e.

$$L[f_1(t)] = F_1(s) \quad \text{and} \quad L[f_2(t)] = F_2(s)$$

then for any number a or b the following is true

$$L[af_1(t) + bf_2(t)] = L[af_1(t)] + L[bf_2(t)] = aL[f_1(t)] + bL[f_2(t)] = aF_1(s) + bF_2(s)$$

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BASIC PROPERTIES OF LAPLACE TRANSFORM DERIVATIVE

If function $f(t)$ and its derivative $f'(t)$ are L-transformable, then derivative transform could be expressed as transform of same function in following way

$$L[f'(t)] = sL[f(t)] - f(0^+) = sF(s) - f(0^+)$$

where $f(0^+)$ - right-hand limit of the function $f(t)$ at point $t = 0$ (initial value of the function $f(t)$)

Transform of the n^{th} -order derivative of function $f(t)$ can be calculated as

$$L[f^{(n)}(t)] = s^n F(s) - \sum_{k=1}^n s^{(n-k)} f^{(k-1)}(0^+)$$

If initial conditions are equal to zero then it shows clearly that differentiating a function in time domain corresponds to multiplication of L-transform of same function by s taken to the power equals to derivative's order.

$$L[f^{(n)}(t)] = s^n F(s)$$

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BASIC PROPERTIES OF LAPLACE TRANSFORM INTEGRAL

If function $f(t)$ is L-transformable, then integral transform can be defined as transform of same function in following way

$$L\left[\int f(t)\right] = \frac{F(s)}{s}$$

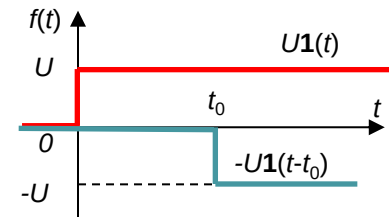
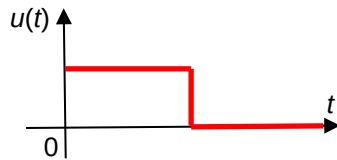
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BASIC PROPERTIES OF LAPLACE TRANSFORM SHIFT IN TIME DOMAIN

If function $f(t)\mathbf{1}(t)$ is L-transformable and has transform $F(s)$, then the transform of the function $f(t-t_0)\mathbf{1}(t-t_0)$ for $t \geq 0$ is defined as

$$L[f(t-t_0)\mathbf{1}(t-t_0)] = F(s)e^{-st_0}$$

Example: A voltage function in the form of a square pulse is given. The transformation of this function is to be found.



$$f(t) = \begin{cases} 0 & \text{for } t \leq 0 \text{ and } t_0 \leq t \\ U & \text{for } 0 < t < t_0 \end{cases}$$

$$f(t) = U\mathbf{1}(t) - U\mathbf{1}(t-t_0)$$

According to the linearity theorem and time shift theorem, we write:

$$F(s) = L[U\mathbf{1}(t) - U\mathbf{1}(t-t_0)] = L[U\mathbf{1}(t)] - L[U\mathbf{1}(t-t_0)] = \frac{U}{s} - \frac{U}{s} e^{-st_0}$$

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BASIC PROPERTIES OF LAPLACE TRANSFORM SHIFT IN COMPLEX DOMAIN

If $F(s)$ is a transform of function $f(t)$ and a is any complex or real number, then the transform with the shifted argument satisfies the following relationship

$$F(s + a) = L[e^{-at} f(t)]$$

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BASIC PROPERTIES OF LAPLACE TRANSFORM TRANSFORM OF A PERIODIC FUNCTION

If $f(t)$ is a periodic function i.e.,

$$f(t) = f(t + kT) \quad k \in \mathbb{Z}$$

Then the corresponding L-transform is equal to:

$$F(s) = \frac{F_T(s)}{1 - e^{-sT}}$$

where $F_T(s)$ is L-transform of a function $f(t)$ within a single period T :

$$F_T(s) = \int_{t_0}^{t_0+T} f(\tau) e^{-s\tau} d\tau$$

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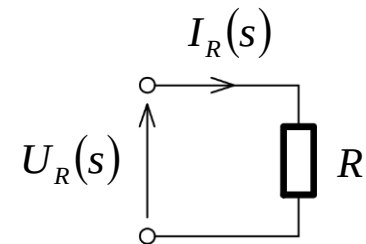
CIRCUIT ELEMENT'S MODEL RESISTOR

The electrical waveforms of the voltage and current of the resistor R are subject to Ohm's law

$$u_R(t) = Ri_R(t)$$

After applying the Laplace transformation and using the linearity we get

$$U_R(s) = RI_R(s)$$



The above formula expresses Ohm's law in operator form. It shows that the operator model of the resistor is characterized by its resistance R

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CIRCUIT ELEMENT'S MODEL INDUCTOR

Time domain

$$i_L(t) = \frac{1}{L} \int u_L(t) dt$$

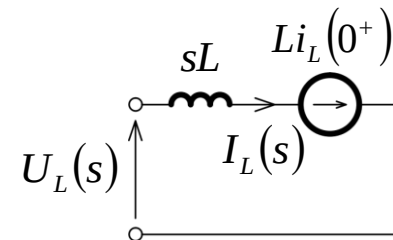
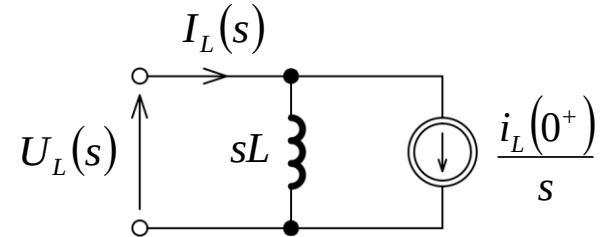
$$u_L(t) = L \frac{di_L(t)}{dt}$$

Laplace domain

$$I_L(s) = \frac{U_L(s)}{sL} + \frac{i_L(0^+)}{s}$$

$$U_L(s) = sLI_L(s) - Li_L(0^+)$$

Element's model



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CIRCUIT ELEMENT'S MODEL CAPACITOR

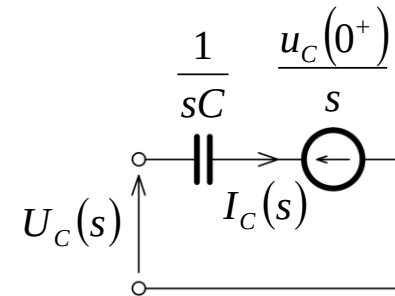
Time domain

$$u_C(t) = \frac{1}{C} \int i_C(t) dt$$

Laplace domain

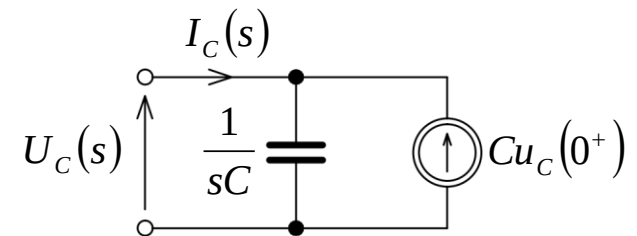
$$U_C(s) = \frac{I_C(s)}{sC} + \frac{u_C(0^+)}{s}$$

Element's model



$$i_C(t) = C \frac{du_C(t)}{dt}$$

$$I_C(s) = sCU_C(s) - Cu_C(0^+)$$



Introduction to Transient Analysis

CALCULATION ALGORITHM

Assumptions:

- the electrical circuit in the time domain is given i.e., its structure (circuit diagram) and parameter values are known,
- supply functions are given ($e(t)$, $j(t)$) i.e., their functional description or graph of variation in time,
- commutation time t_k is given explicitly e.g., $t_k = 0$ s,
- before commutation (for $t \rightarrow t_k$) the circuit is in an explicitly described steady state

Algorithm stages:

1. Derive initial conditions based on steady state analysis before commutation (for $t \rightarrow t_k$) according to commutation law
2. Calculate operator forms of supply signals ($E(s)$, $J(s)$)
3. Create circuit diagram in Laplace domain, taking into account initial conditions
4. Determine desired signal/signals in operator form
5. Do inverse Laplace transform (transform table, linearity theorem, partial fraction decomposition), draw a graph showing the signal/signals in time domain (before and after commutation)

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1st-ORDER CIRCUIT – EXAMPLE 1

At $t = 0$, the switch W was opened. Draw the current of the inductor L before and after this moment.

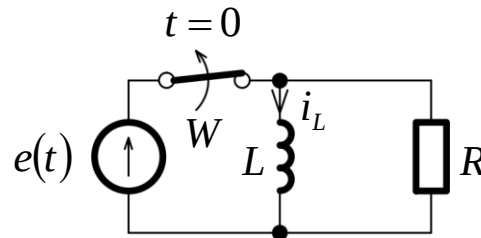
Data:

for $t < 0$ the system was in a steady state,

$$L = 0.1 \text{ H}$$

$$R = 10 \, \Omega$$

$$e(t) = 157\sin(314t+120^\circ)$$



Introduction to Transient Analysis

1st-ORDER CIRCUIT – EXAMPLE 1

1. We determine the initial conditions (I.C.) based on the knowledge of the state of the circuit for $t < 0$ and the commutation laws:

Complex phasors:

$$E = 157e^{j120^\circ}$$
$$I_L = \frac{E}{j\omega L} = \frac{157e^{j120^\circ}}{34.1e^{j90^\circ}} = 5e^{j30^\circ}$$

Time domain for steady state:

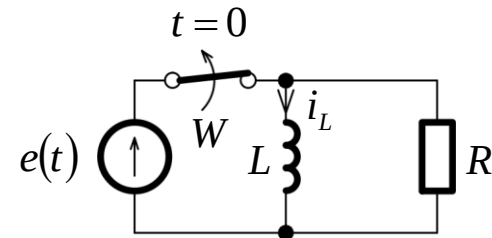
$$i_L(t) = 5\sin(314t + 30^\circ)[A]$$

Time domain for commutation time:

$$i_L(t = 0) = 5\sin(30^\circ) = 2.5[A]$$

According to commutation law:

$$i_L(0^+) = i_L(0^-) = 2.5[A]$$



Introduction to Transient Analysis

1st-ORDER CIRCUIT – EXAMPLE 1

1. We draw new circuit diagram taking into account initial conditions:
2. We analyse the operating circuit and determine the operating form of the desired signal.

$$sLI_L(s) + RI_L(s) = Li_L(0^+)$$

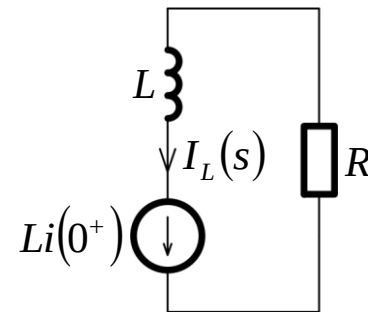
$$I_L(s)(sL + R) = Li_L(0^+)$$

$$I_L(s) = \frac{Li_L(0^+)}{sL + R}$$

$$I_L(s) = i_L(0^+) \frac{1}{s + \frac{R}{L}} \quad \longrightarrow$$

$$F(s) = \frac{1}{s + a} \quad \longleftrightarrow \quad f(t) = 1e^{-at}$$

from Laplace transforms table



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1st-ORDER CIRCUIT – EXAMPLE 1

Time constant τ :

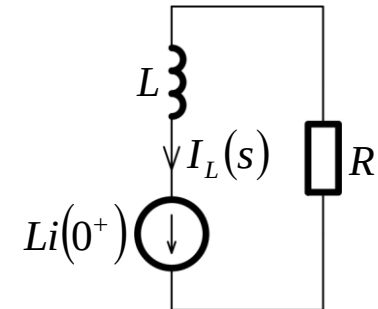
$$I_L(s) = i_L(0^+) \frac{1}{s + \frac{R}{L}} \Rightarrow \boxed{F(s) = \frac{1}{s + a} \iff f(t) = \mathbf{1}e^{-at}}$$

from Laplace transforms table

$$i_L(t) = \mathbf{1}i_L(0^+)e^{-at} = \mathbf{1}i_L(0^+)e^{-\frac{R}{L}t} = \mathbf{1}i_L(0^+)e^{-\frac{t}{\tau}}$$

where:

$$\tau = \frac{L}{R} [s]$$



Introduction to Transient Analysis

1st-ORDER CIRCUIT – EXAMPLE 1

Physical meaning of the constant τ :

$$f(t) = 1e^{-\frac{t}{\tau}}$$

Just after commutation ($t = 0^+$):

$$f(0) = 1$$

In steady state ($t \rightarrow \infty$):

$$\lim_{t \rightarrow \infty} f(t) = \lim_{t \rightarrow \infty} \left(1e^{-\frac{t}{\tau}} \right) = 0$$

The constant τ defines how fast is the circuit's response:

for	$t = \tau$	$e^{-\frac{t}{\tau}} = 0.3679$	63.21%
for	$t = 2\tau$	$e^{-\frac{t}{2\tau}} = 0.1353$	86.47%
for	$t = 3\tau$	$e^{-\frac{t}{3\tau}} = 0.0498$	95.02%
for	$t = 4\tau$	$e^{-\frac{t}{4\tau}} = 0.0183$	98.17%
for	$t = 5\tau$	$e^{-\frac{t}{5\tau}} = 0.0067$	99.3%

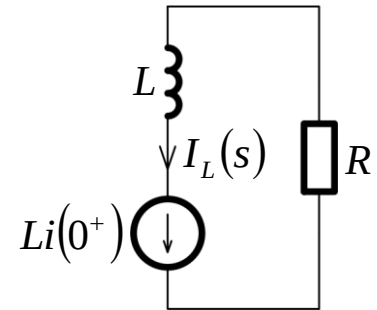
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1st-ORDER CIRCUIT – EXAMPLE 1

Time constant τ :

$$i_L(t) = 1 \cdot 2.5e^{-100t} = 1 \cdot 2.5e^{-\frac{t}{0.01}}$$

$$\tau = \frac{L}{R} = 10[ms]$$



Time constant τ can be used to determine slope of the exponential response at $t = 0$

$$\frac{df(t)}{dt} = \frac{d\left(Ae^{-\frac{t}{\tau}}\right)}{dt} = -\frac{A}{\tau}e^{-\frac{t}{\tau}} \Big|_{t=0} = -\frac{A}{\tau}$$

by creating a linear function that passes through point $(0, i_L(0^+))$ and passes through time axis (for $t = \tau$):

$$y = -\frac{A}{\tau}t + A$$

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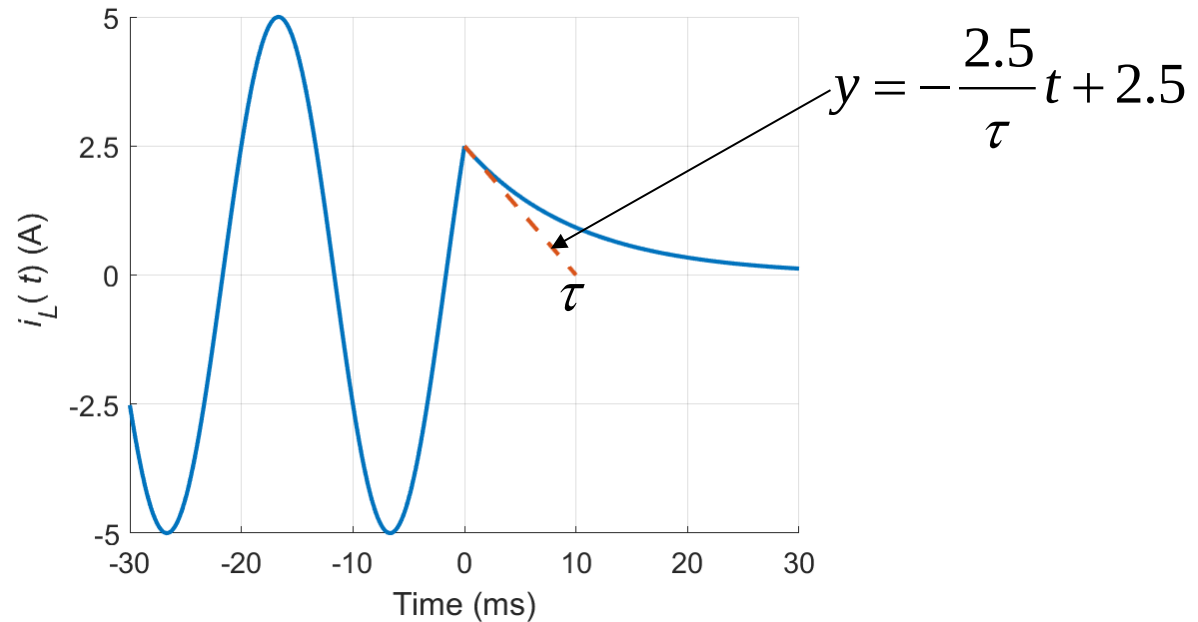
1st-ORDER CIRCUIT – EXAMPLE 1

Steady state ($t < 0$):

$$i_L(t) = 5\sin(314t + 30^\circ)[A]$$

Dynamic state ($t > 0$):

$$i_L(t) = 1 \cdot 2.5e^{-100t} = 1 \cdot 2.5e^{-\frac{t}{0.01}}[A]$$

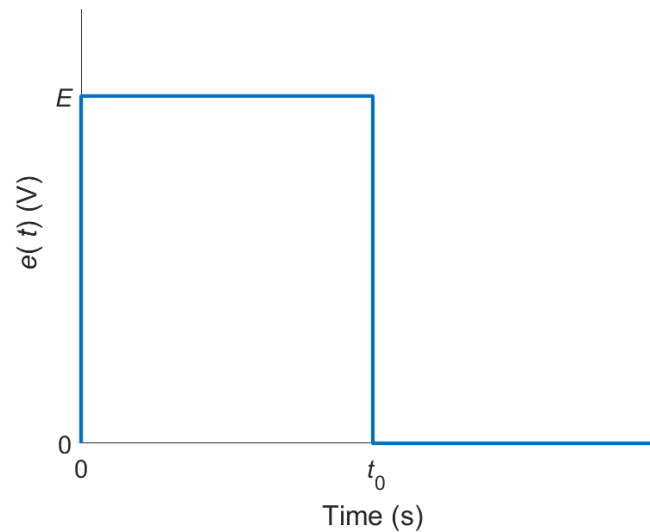
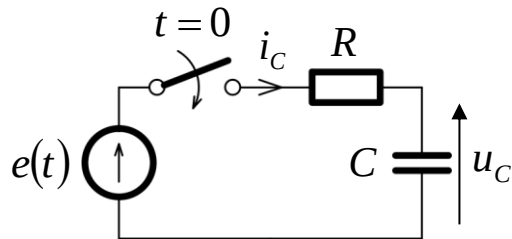


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1st-ORDER CIRCUIT – EXAMPLE 2

At $t = 0$, the switch was closed. Draw current and voltage of the capacitor C.

Data:



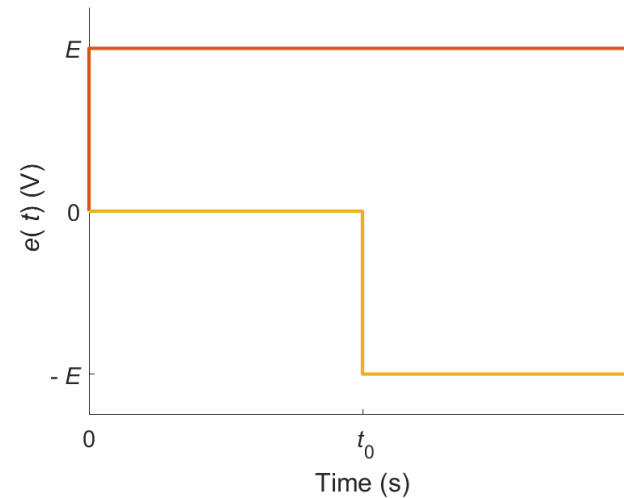
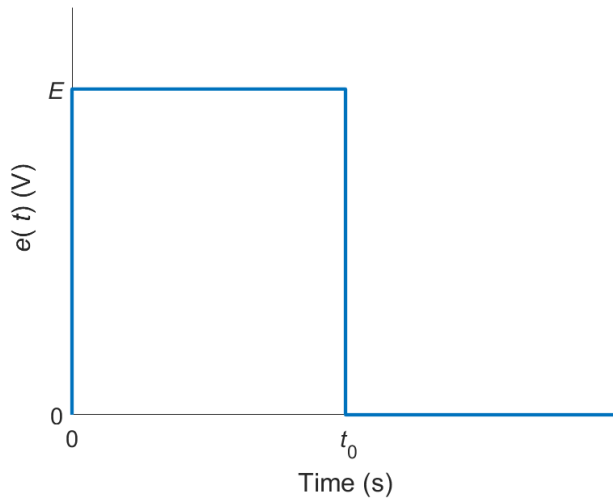
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1st-ORDER CIRCUIT – EXAMPLE 2

1. We determine initial conditions (I.C.) based on the knowledge of the circuit's state for $t < t_k$ and commutation laws:

$$u_c(0^+) = u_c(0^-) = 0$$

2. We determine supply signals in Laplace domain:



$$e(t) = E \cdot \mathbf{1}(t) - E \cdot \mathbf{1}(t - t_0)$$

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1st-ORDER CIRCUIT – EXAMPLE 2

$$e(t) = E \cdot \mathbf{1}(t) - E \cdot \mathbf{1}(t - t_0)$$

We know that:

$$\mathcal{L}[E \cdot \mathbf{1}(t)] = \frac{E}{s}$$

According to the theorem on shift in time domain:

$$\text{if } \mathcal{L}[f(t)] = F(s) \text{ then } \mathcal{L}[f(t - t_0) \cdot (t - t_0)] = F(s)e^{-st_0}$$

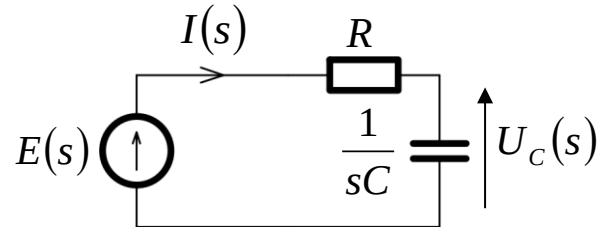
and linearity theorem we can write:

$$E(s) = \frac{E}{s} - \frac{E}{s}e^{-st_0} = \frac{E}{s}(1 - e^{-st_0})$$

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1st-ORDER CIRCUIT – EXAMPLE 2

3. We draw the circuit diagram taking into account I.C.



4. We analyse the circuit and determine the operator form of the desired signal.

The circuit current in operator form:

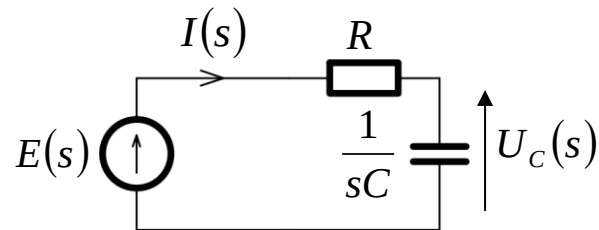
$$\begin{aligned} I(s) &= \frac{E(s)}{R + \frac{1}{sC}} = \frac{s E(s)}{s R + \frac{1}{C}} = \frac{s E(s)}{R \left(s + \frac{1}{RC} \right)} = \frac{s \frac{E}{s} (1 - e^{-st_0})}{R \left(s + \frac{1}{RC} \right)} = \\ &= \frac{E}{R} \frac{1}{s + \frac{1}{RC}} (1 - e^{-st_0}) = \frac{E}{R} \left[\frac{1}{s + \frac{1}{RC}} - \frac{1}{s + \frac{1}{RC}} e^{-st_0} \right] \end{aligned}$$

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1st-ORDER CIRCUIT – EXAMPLE 2

Operator voltage across the capacitor:

$$\begin{aligned} U_c(s) &= I(s) \frac{1}{sC} = \frac{1}{sC} \frac{E}{R} \left[\frac{1}{s + \frac{1}{RC}} - \frac{1}{s + \frac{1}{RC}} e^{-st_0} \right] = \\ &= \frac{E}{RC} \left[\frac{1}{s \left(s + \frac{1}{RC} \right)} - \frac{1}{s \left(s + \frac{1}{RC} \right)} e^{-st_0} \right] \end{aligned}$$



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1st-ORDER CIRCUIT – EXAMPLE 2

5. We find the signal in time domain:

$$I(s) = \frac{E}{R} \left[\frac{1}{s + \frac{1}{RC}} - \frac{1}{s + \frac{1}{RC}} e^{-st_0} \right]$$

$$f(t) = \mathbf{1}e^{-at} \longleftrightarrow F(s) = \frac{1}{s + a}$$

from Laplace transforms table

Hence:

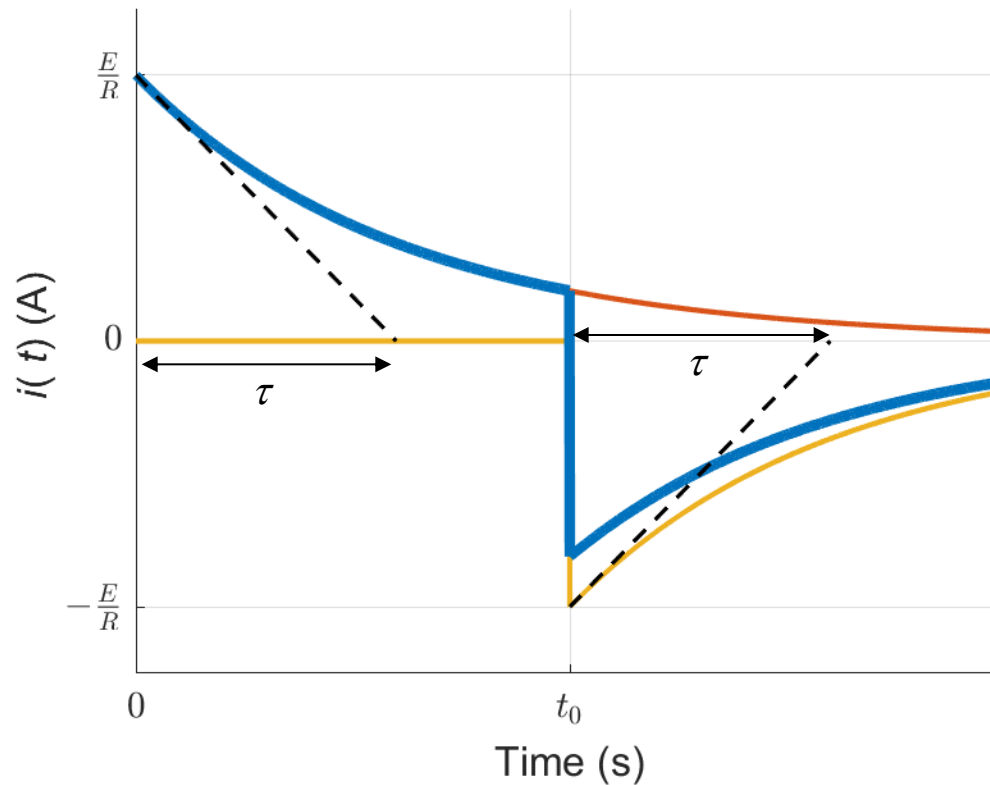
$$i(t) = \frac{E}{R} \left[e^{-\frac{1}{RC}t} \mathbf{1}(t) - e^{-\frac{1}{RC}(t-t_0)} \mathbf{1}(t - t_0) \right]$$

$$i(t) = \frac{E}{R} e^{-\frac{1}{RC}t} \mathbf{1}(t) - \frac{E}{R} e^{-\frac{1}{RC}(t-t_0)} \mathbf{1}(t - t_0)$$

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1st-ORDER CIRCUIT – EXAMPLE 2

$$i(t) = \frac{E}{R} e^{-\frac{1}{RC}t} \mathbf{1}(t) - \frac{E}{R} e^{-\frac{1}{RC}(t-t_0)} \mathbf{1}(t-t_0)$$



$$\tau = RC$$

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1st-ORDER CIRCUIT – EXAMPLE 2

$$U_c(s) = \frac{E}{RC} \left[\frac{1}{s \left(s + \frac{1}{RC} \right)} - \frac{1}{s \left(s + \frac{1}{RC} \right)} e^{-st_0} \right]$$

$$f(t) = \mathbf{1} \frac{1}{a} (1 - e^{-at}) \longleftrightarrow F(s) = \frac{1}{s(s+a)}$$

from Laplace transforms table

Hence:

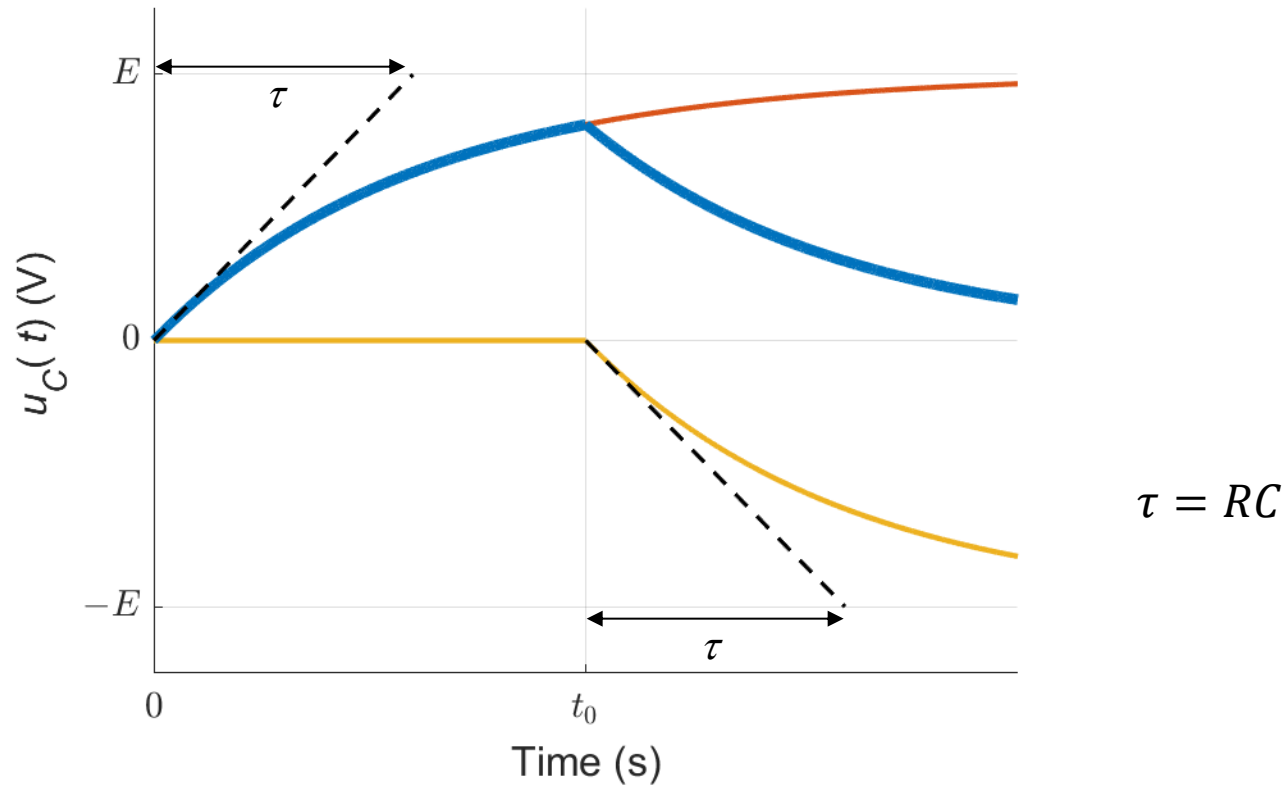
$$u_c(t) = \frac{E}{RC} \left[RC(1 - e^{-\frac{1}{RC}t}) \mathbf{1}(t) - RC(1 - e^{-\frac{1}{RC}(t-t_0)}) \mathbf{1}(t - t_0) \right] =$$

$$u_c(t) = E(1 - e^{-\frac{1}{RC}t}) \mathbf{1}(t) - E(1 - e^{-\frac{1}{RC}(t-t_0)}) \mathbf{1}(t - t_0)$$

Introduction to Transient Analysis

1st-ORDER CIRCUIT – EXAMPLE 2

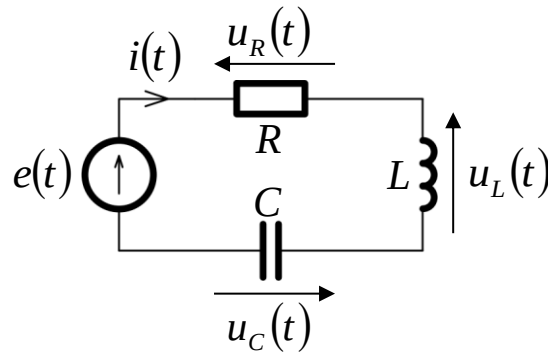
$$u_c(t) = E\left(1 - e^{-\frac{1}{RC}t}\right)\mathbf{1}(t) - E\left(1 - e^{-\frac{1}{RC}(t-t_0)}\right)\mathbf{1}(t - t_0)$$



Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

One of the simplest representatives of such circuits is the one shown below:

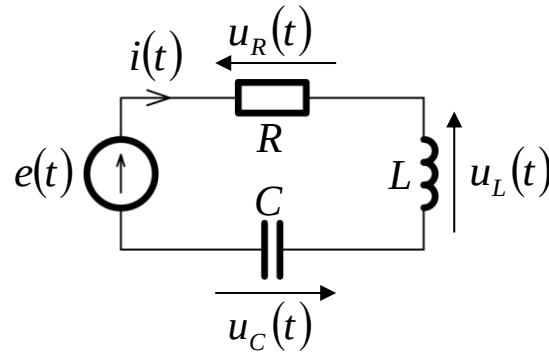


Let's assume that the supplying voltage of such a circuit is a signal described with the unit step function $e(t) = E1(t)$. Let's assume that we are looking for the current function $i(t)$.

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

1. We determine the initial conditions (I.C.) based on the knowledge of the state of the circuit for $t < t_k$ and the commutation laws:



Due to the fact that for $t < 0$ $E = 0$, initial conditions in accordance with I and II law of commutation can be written as:

$$\left. \begin{aligned} i_L(0^-) &= i_L(0^+) = 0 \\ u_C(0^-) &= u_C(0^+) = 0 \end{aligned} \right\}$$

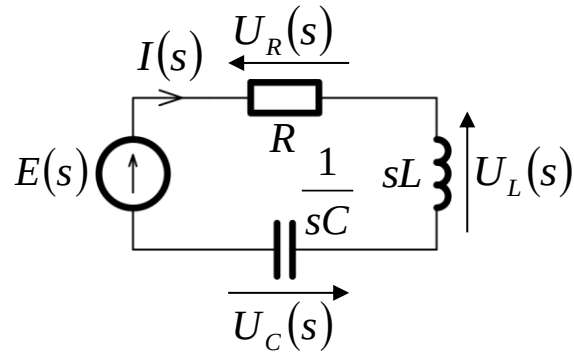
2. We determine the supplying voltage in operator domain:

$$E(s) = \mathcal{L}[e(t)] = \mathcal{L}[E] = \frac{E}{s}$$

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

3. We draw the circuit diagram taking into account I.C.



4. We analyse the circuit and determine operator form of the desired signal:

$$I(s) = \frac{E(s)}{Z(s)} = \frac{\frac{E}{s}}{R + sL + \frac{1}{sC}} = \frac{E}{L} \cdot \frac{1}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

5. We find the signal in time domain.

The equation describing the considered current is a rational function

$$I(s) = \frac{E}{L} \cdot \frac{L(s)}{M(s)}$$

where

$$L(s) = \frac{E}{L}$$

$$M(s) = s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

By solving the above equation we obtain the roots of the current function:

$$\left. \begin{aligned} s_1 &= \frac{-\frac{R}{L} + \sqrt{\Delta}}{2} \\ s_2 &= \frac{-\frac{R}{L} - \sqrt{\Delta}}{2} \end{aligned} \right\}$$

$$\Delta = \left(\frac{R}{L}\right)^2 - \frac{4}{LC}$$

is the differentiator $M(s)$

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

There are three possible solutions:

A) $\Delta > 0$ – two real roots of $M(s)$ if $R > 2\sqrt{\frac{L}{C}}$

B) $\Delta = 0$ – one double root of $M(s)$ if $R = 2\sqrt{\frac{L}{C}}$

C) $\Delta < 0$ – two complex-conjugated roots of $M(s)$ if $R < 2\sqrt{\frac{L}{C}}$

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

Case A – OVERDAMPED RESPONSE

Both roots are real:

$$\left. \begin{aligned} s_1 &= -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \\ s_2 &= -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \end{aligned} \right\} \quad R > 2\sqrt{\frac{L}{C}}$$

In time domain the current takes the form:

$$i_A(t) = \frac{E}{2L \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}} [e^{s_1 t} - e^{s_2 t}]$$

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

Case B – CRITICALLY DAMPED RESPONSE

One double root:

$$s_1 = s_2 = -\frac{R}{2L}$$

$$R = 2\sqrt{\frac{L}{C}}$$

In time domain the current takes the form:

$$i_B(t) = \frac{E}{L} t e^{-\frac{R}{2L}t}$$

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

Case C – UNDERDAMPED RESPONSE (OSCILLATIONS)

Two complex-conjugated roots:

$$\left. \begin{aligned} s_1 &= -\frac{R}{2L} + j \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} \\ s_2 &= -\frac{R}{2L} - j \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} \end{aligned} \right\} R < 2 \sqrt{\frac{L}{C}}$$

In time domain the current takes the form:

$$i_C(t) = \frac{E}{\omega L} e^{-\frac{R}{2L}t} \sin(\omega t)$$

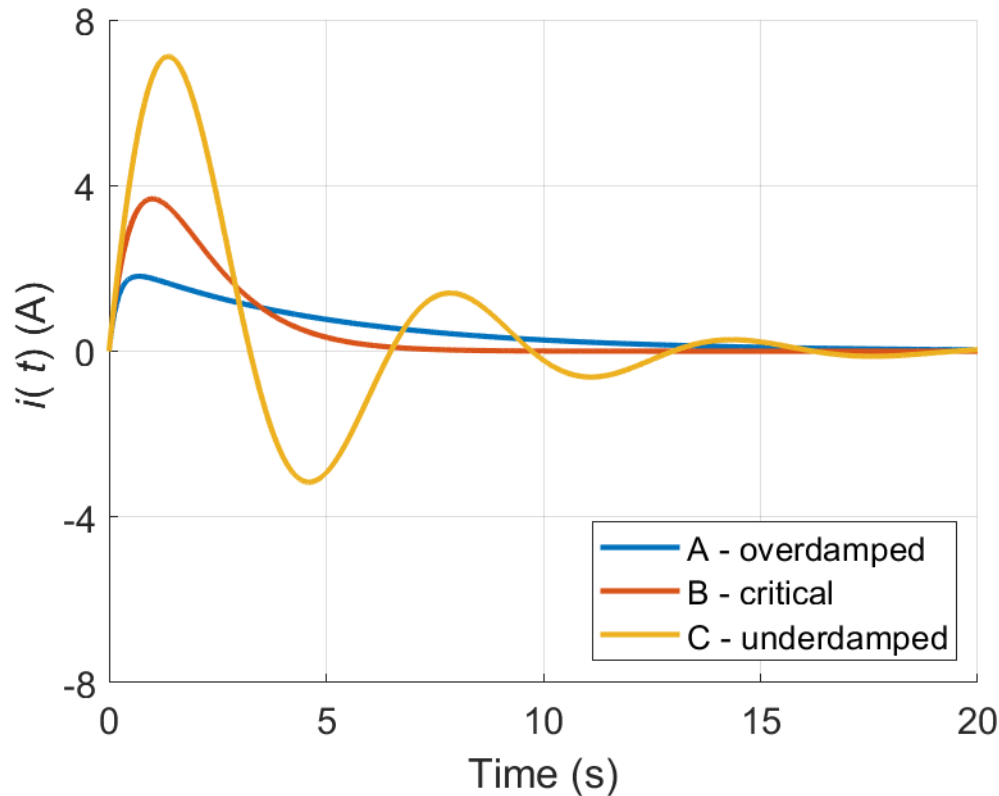
where

$$\omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

Introduction to Transient Analysis

2nd-ORDER CIRCUIT – EXAMPLE

EXEMPLARY RESPONSES



Data:

$$E = 10 \text{ V}$$

$$R_A = 5 \text{ } \Omega$$

$$R_B = 2 \text{ } \Omega$$

$$R_C = 0.5 \text{ } \Omega$$

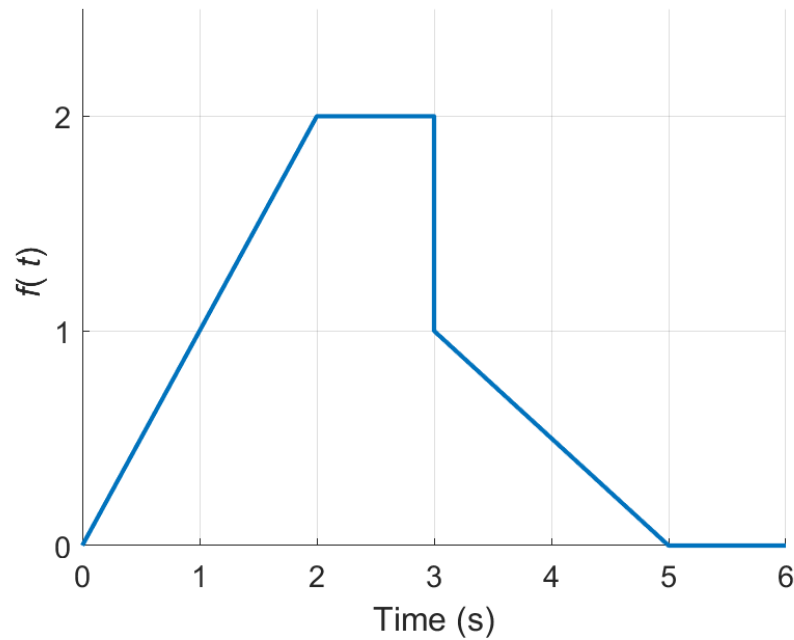
$$L = 1 \text{ H}$$

$$C = 1 \text{ F}$$

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TRANSFORMS OF ARBITRARY SIGNALS

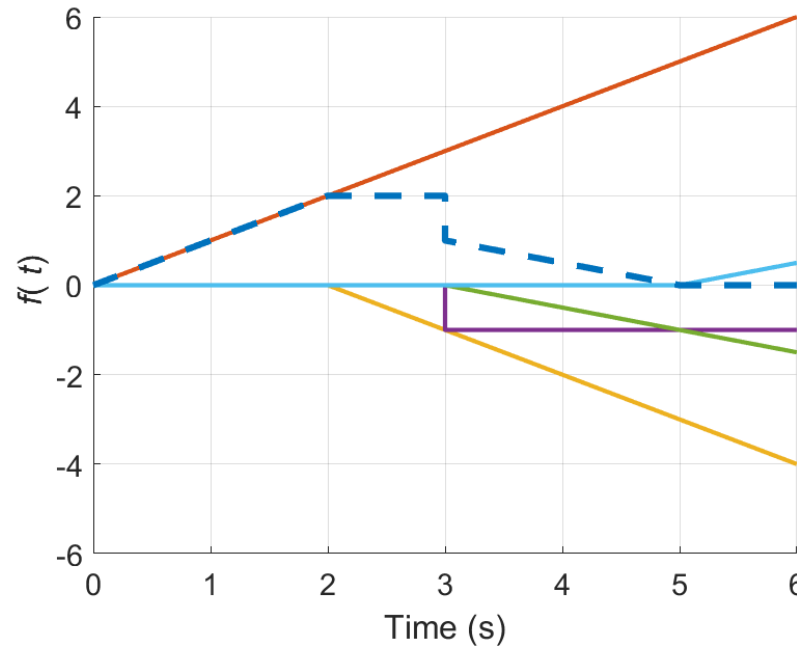
EXAMPLE 1



Introduction to Transient Analysis

TRANSFORMS OF ARBITRARY SIGNALS

EXAMPLE 1



The above signal can be described as:

$$f(t) = t \cdot 1(t) - (t-2) \cdot 1(t-2) - 1(t-3) - \frac{1}{2}(t-3) \cdot 1(t-3) + \frac{1}{2}(t-5) \cdot 1(t-5)$$

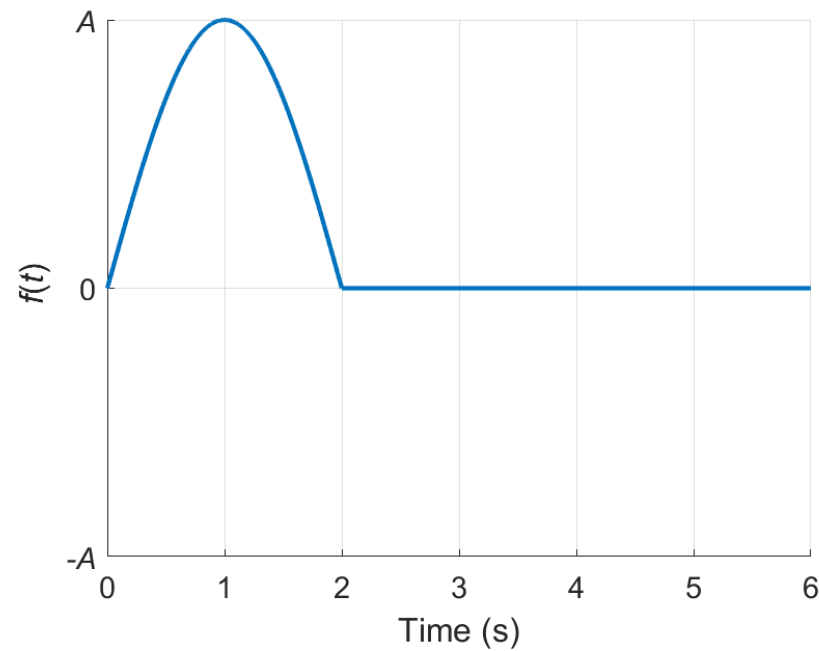
Laplace transform for this function is equal:

$$F(s) = \frac{1 - e^{-2s} - s e^{-3s} - \frac{1}{2} e^{-3s} + \frac{1}{2} e^{-5s}}{s^2}$$

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TRANSFORMS OF ARBITRARY SIGNALS

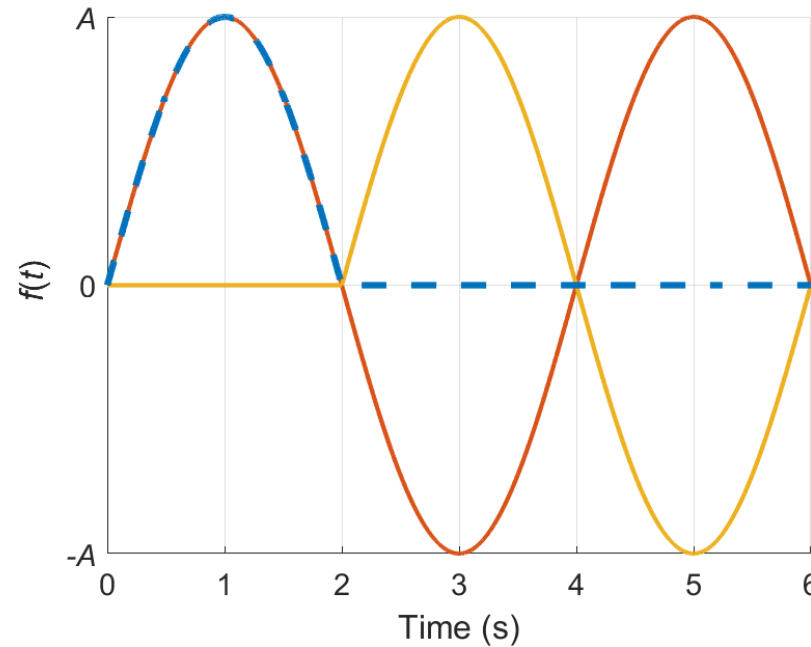
EXAMPLE 2



Introduction to Transient Analysis

TRANSFORMS OF ARBITRARY SIGNALS

EXAMPLE 2



The above signal can be described as:

$$f(t) = A \left[\sin \frac{\pi}{T} t \mathbf{1}(t) + \sin \frac{\pi}{T} (t - T) \mathbf{1}(t - T) \right]$$

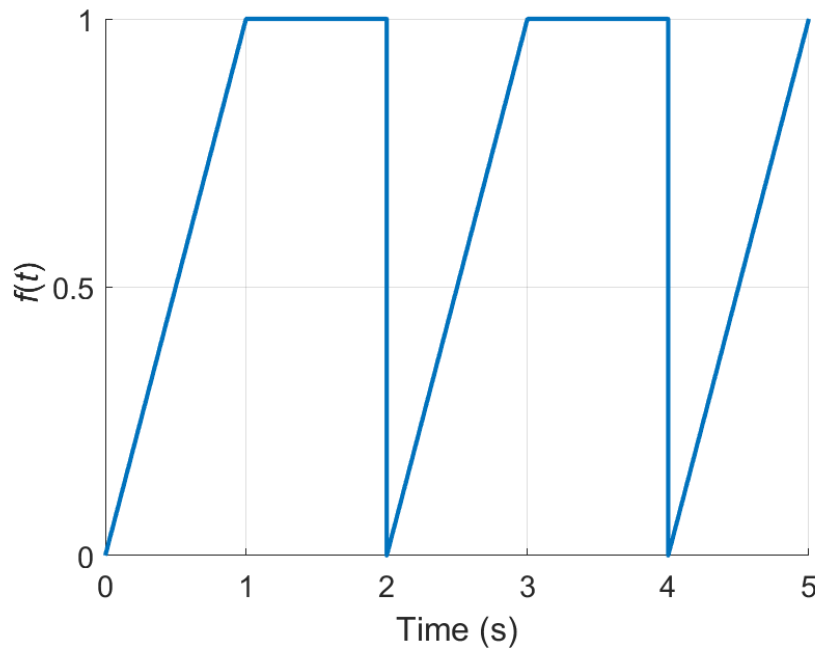
Laplace transform for this function is equal:

$$F(s) = \frac{A \frac{\pi}{T} (1 + e^{-sT})}{s^2 + \left(\frac{\pi}{T}\right)^2}$$

Introduction to Transient Analysis

TRANSFORMS OF ARBITRARY SIGNALS

EXAMPLE 3 – PERIODIC SIGNAL



The transform of a periodic function with period T i.e., such that:

$$f(t) = f(t - kT), k = 1, 2, \dots,$$

has the form of:

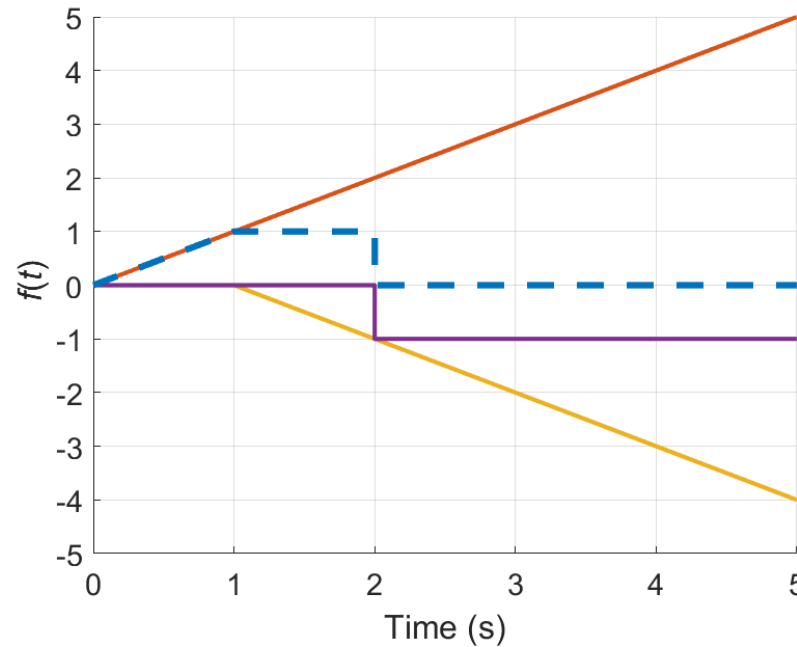
$$F(s) = \frac{F_T(s)}{1 - e^{-sT}},$$

where $F_T(s)$ is transform of the pulse $f_T(t)$, which is equal to the function $f(t)$ in the interval $\langle 0, T \rangle$ and zero outside that interval

Introduction to Transient Analysis

TRANSFORMS OF ARBITRARY SIGNALS

EXAMPLE 3



In this example $T = 2$, and:

$$f_T(t) = t\mathbf{1}(t) - (t-1)\mathbf{1}(t-1) - \mathbf{1}(t-2).$$

After determining the transform, we get:

$$F_T(s) = \frac{1 - e^{-s} - se^{-2s}}{s^2}, \text{ which leads to } F(s) = \frac{1 - e^{-s} - se^{-2s}}{s^2(1 - e^{-2s})}$$

Introduction to Transient Analysis

INVERSE TRANSFORM OF ARBITRARY SIGNALS

PARTIAL FRACTION DECOMPOSITION

Find the inverse Laplace transform of a function

$$F(s) = \frac{s^3 + 16s^2 + 33s + 42}{s(s + 1)(s + 3)(s + 7)}$$

Function $F(s)$ is rational function. Moreover degree of the numerator is smaller than degree of the denominator and roots of the denominator are single. We can decompose that function to partial fractions i.e., write it as a sum:

$$F(s) = \frac{c_1}{s} + \frac{c_2}{s + 1} + \frac{c_3}{s + 3} + \frac{c_4}{s + 7}$$

Introduction to Transient Analysis

INVERSE TRANSFORM OF ARBITRARY SIGNALS

PARTIAL FRACTION DECOMPOSITION

$$F(s) = \frac{s^3 + 16s^2 + 33s + 42}{s(s+1)(s+3)(s+7)}$$

We can calculate coefficients of c_k as shown:

$$c_1 = sF(s) \Big|_{s=0} = \frac{s^3 + 16s^2 + 33s + 42}{(s+1)(s+3)(s+7)} \Big|_{s=0} = 2$$

$$c_2 = (s+1)F(s) \Big|_{s=-1} = \frac{s^3 + 16s^2 + 33s + 42}{s(s+3)(s+7)} \Big|_{s=-1} = -2$$

$$c_3 = (s+3)F(s) \Big|_{s=-3} = \frac{s^3 + 16s^2 + 33s + 42}{s(s+1)(s+7)} \Big|_{s=-3} = \frac{5}{2}$$

$$c_4 = (s+7)F(s) \Big|_{s=-7} = \frac{s^3 + 16s^2 + 33s + 42}{s(s+1)(s+3)} \Big|_{s=-7} = -\frac{3}{2}$$

Introduction to Transient Analysis

INVERSE TRANSFORM OF ARBITRARY SIGNALS

PARTIAL FRACTION DECOMPOSITION

We can write function $F(s)$ as:

$$F(s) = \frac{2}{s} - \frac{2}{s+1} + \frac{\frac{5}{2}}{s+3} - \frac{\frac{3}{2}}{s+7}$$

Inverse transforms of those individual components are easy to determine with the use of the presented table. Finally, we get:

$$f(t) = \left(2 - 2e^{-t} + \frac{5}{2}e^{-3t} - \frac{3}{2}e^{-7t} \right) \mathbf{1}(t).$$